

Physical Climate Risk Methodology

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This document describes the OS-Climate physical climate risk methodology, although additional implementation detail is found in the [physrisk library documentation](#) – along with code-library-specific topics. This is done where it is more convenient to document certain details via runnable Jupyter notebooks which support interactive features. Sources of hazard and vulnerability functions are examples of this.

1 Introduction

The changing climate introduces new risks. These can be grouped into:

1. Physical risks – risks arising from the physical effects of climate change,
2. Transition risks – risks arising from the transition to a low-carbon economy,
3. Liability risks – considered a third by some [43], these are the risks arising when those affected by anthropogenic climate change seek compensation.

The methodology presented in this document concerns the assessment of physical risk. Physical risk comes from changes in climate *hazards*. A hazard is the potential occurrence of a climate-related physical phenomenon that can impact human and ecological systems [31, 43, 21]. More precisely, the impact may be loss of life, injury, or other health impact, as well as damage and loss to property, infrastructure, livelihoods, service provision, ecosystems, and environmental resources (see Annex II of [27]). Hazards can be divided into *acute hazards* and *chronic hazards*. An acute hazard is the potential occurrence of an *event*, for example a heat wave, inundation (flood) or hurricane. A chronic hazard is the potential occurrence of a trend in climate parameters such as an increase in average temperature, sea-level or water stress indices.

The authors of [29] argue that sudden accute events are the most likely to generate ‘*material shocks to the financial sector in the near-term*’ and note that techniques to generate probabilistic scenarios to inform financial decision making is well-developed in the insurance industry. This is taken as a guiding principle, that a methodology for the measurement of physical climate risk should be adapted from the techniques developed by catastrophe modellers. The authors also note that ‘*the financial risks from*

*physical climate shocks cannot be approximated by considering only average annual costs of weather extremes, even on long timescales. Larger, rarer events can cause significant damage and disruption and have long-lived impacts.*¹ Another guiding principle is this handling of rare events.

A model designed to quantify physical risk must take into account: a) hazard likelihood of occurrence b) the damage or disruption caused c) the consequence of this damage/disruption. Damage/disruption caused by a hazard is determined by the *vulnerability* of the asset that is exposed¹. With a focus on financial risk, damage/disruption refers to damage of financial assets and disruption to business activities. More generally, damage can refer to natural assets and disruption to populations and ecosystems. As an example, the physical infrastructure of a power generating asset may be damaged by inundation and its electricity production may be disrupted, leading to a loss in revenue.

We assign explicit names to these three components, a), b) and c) for the financial risk case:

- a) Hazard
- b) Vulnerability
- c) Financial

The authors of [2] note that, *‘historical damage data may be insufficient or even irrelevant for generating a risk assessment that reflects the current or future risk reality’* and further that *‘physical climate risk assessments now increasingly use data sets that can individually characterize hazard, exposure, and vulnerability’*. The methodology presented in this document is indeed intended to perform such assessments.

A precise model of the physical risk from a single (localized) asset or business activity must consider a hazard’s likelihood of occurrence *at the asset’s locale* (i.e. if the asset is exposed to the hazard) and vulnerability to the hazard particular to that asset. This requires specific knowledge of the asset. For example a power station that relies on air-cooling might be disrupted by a period of extremely high air temperature. In addition, an asset may be impacted through its reliance on other assets; a manufacturing facility may rely on continuity of electricity supply for example.

Such precise models of physical risk can be complex and may rely on information that is not readily available. For these reasons, approximations

¹ *Exposure* of an asset to a hazard is defined after [17]; for most purposes this is determined by asset location.

are commonly used, although approximate models still typically include hazard, vulnerability and financial components [4, 43] – even global-scale impact analyses intended to be used in macroeconomic models.

The purpose of this paper is to present the methodology of a framework that is sufficiently generic to be used for a wide range of physical climate risk models, both precise and approximate as required. The ability to perform precise, fine-grained calculations is an important requirement therefore. This paper serves as a specification for use in the ‘*physrisk*’ OS-Climate (OS-C) [24] physical climate risk calculation module.

OS-C aims to provide a platform unconstrained by any one particular methodology choice, but takes inspiration from natural catastrophe modelling [21] and in particular the *Oasis Loss Modelling Framework* [23] (henceforth *Oasis LMF*), which was designed to accommodate a wide range of catastrophe models and analyse physical risk in the context of the insurance market. Similarly to *Oasis LMF*, the library adopts a modular approach. This approach allows the user to change easily a particular modelling method, whilst maintaining the integration of the components.

In the following, models of hazards, vulnerability and financial impact are discussed in more detail. In a later section these are presented more formally.

Hazard. As noted above, hazard models come in two varieties: models of acute hazards – events – and models of chronic hazards – long-term shifts in climate parameters. In climate risk events are *climate-conditioned*: based not just on historical events but also future projections under different assumptions.

A. Accute hazard models.

Accute hazard models may be *event-based* or *return-period-based*. *Event-based hazard models* are models of individual events and are common in natural catastrophe modelling. Typically, for a large number of events, a model provides spatial distributions (i.e. a map) of the probabilities of occurrence of different event intensities. These distributions are sometimes called ‘hazard footprints’ [37] or ‘hazard maps’. As an example, in the case of inundation, the hazard footprint would provide for different locations the probability of occurrence of different inundation depths – associated with one particular inundation event.

Event-based hazard models are important when the *correlation* of hazards to which assets are exposed is material to the analysis being

performed. For example, if one house on a street is exposed to an inundation it is likely that the house two doors down will also be exposed. This is captured by event-based hazard models: both houses might well appear in the same hazard footprint. The models are therefore important in reinsurance risk calculations, say, where the ability to calculate accurately ‘worst-case’ losses is highly desirable.

Return-period-based hazard models, in contrast, provide at each location a curve of return-periods and corresponding intensities of an acute hazard. For example an inundation model may specify that at a certain latitude and longitude an inundation event with a depth of 50 cm will occur with a return period of 100 years, 70 cm with a return period of 200 years and so on². Such a return-period map provides no correlation information.

Return-period-based approaches offer a significant computational advantage at the expense of the missing correlation information. Appropriate use-cases are:

- when correlation of hazards is not relevant to the calculation, for example when modelling a single asset, or for certain calculations, such as annual average ground-up loss³, or
- when it is satisfactory to model correlation using a heuristic approach (e.g. when obtaining a rough estimate based on 0% or 100% correlation or some estimated average spatial correlation). This may be used in cases where it is desirable to separate low-risk from high risk lending portfolios for example.

B. *Chronic hazards*. As discussed above, acute hazard models provide probability distributions for future events. In contrast chronic hazard models provide climate parameters only. The models are therefore non-probabilistic for a given scenario. For example, a model of average surface temperature would provide values for different latitudes and longitudes under a particular climate scenario (e.g. particular social-economic and representative concentration pathway). *Under this particular scenario*, the assumption is that the climate parameter is certain to take the value.

Hazard likelihood of occurrence is generally scenario-based [4]. That is, hazards are modelled in a historical baseline and then again assuming a particular climate scenario.

² For inundation models, 10 return periods for each location may be specified.

³ For calculation of certain measures only marginal probability of occurrence of a hazard at a given location is relevant. This is discussed in more detail later, but may be true for certain average measures; for measures of tail-risk, in contrast, joint probability is needed for the calculation.

Vulnerability. The vulnerability component measures the potential impact of a catastrophic event on an asset. One very important feature of the methodological approach is that like acute hazard models *vulnerability models are probabilistic*. This is because:

- The damage or disruption to an asset caused by some precisely known event (e.g. flood of known intensity) may not itself be precisely known or knowable. That is, there is an *aleatory uncertainty* [21].
- In many cases the properties of the asset sufficient to ascertain its vulnerability are, although in principle knowable, unknown. That is there is an *epistemic uncertainty* [21].

Both these sources of uncertainty may be material to the calculation in hand and neglecting these is a potential source of model risk⁴. In many climate risk calculations, epistemic uncertainty is particularly important: the vulnerability of assets in some portfolios may be uncertain but this does not prevent, say, the estimation of a likely upper bound to damage or disruption. Such a bound may be wrong however, if one does not allow for the possibility that the asset’s nature is such that it may be significantly more vulnerable than some average level.

At time of writing, physical risk calculations may make use of ‘bulk-assessment’ approaches where accurate asset vulnerability information is unavailable and approximations are therefore required. The modelling framework aims to accommodate bulk-assessment-type models as well as approaches capable of modelling vulnerability more precisely, the assumption being that there is potentially great value in the results obtained from very simple models, as long as the model uncertainty is properly-quantified. The aim is to be able to accommodate both simple and complex models in combination.

Financial. Finally, the financial component is concerned with translating a probability distribution of asset damage/disruption to a loss of profitability for a company, or a loss of value for a lender, insurer, equity stakeholder etc. Clearly, the models used in the Financial module answer questions specific to a certain user: is the ultimate objective that of measuring the physical risk for the company or for one of the asset’s insurer? Depending on the answer to that question, a different financial model might be needed.

⁴ We define model risk to be the risk that the model outputs lead to poor management information or cause bad decisions to be made. Note that model risk is *not* the risk that a model is wrong: all models are in some sense ‘wrong’, although some may be useful.

One important feature of financial models is the role played by insurance. For example, for a lender, what is the impact of climate change on the level of risk of a commercial real estate portfolio? An increase in the frequency and severity of acute hazards may affect the value of the loan collateral but this can depend strongly on whether damage/disruption is covered by insurance and premiums remain low for example.

Design goal The design goal of the ‘*physrisk*’ library is to facilitate the analysis of physical risk from a variety of perspectives. There is no specific market (e.g. insurance market) in mind; the intent is to be sufficiently general to allow any stakeholder to analyse their physical risks. In addition, the framework should introduce as few constraints as possible: the clear hypothesis is that modules are dependent from one another only through their potential input-output relationships. Other than that, the intent is to allow users to have a flexible access to a wide choice of models within each module: the models are only required to have a predictable external behaviour, while the details of the internal workings can be defined with flexibility.

A number of teams have developed models and tools that tackle one specific aspect of the problem. However, it is not currently easy to seamlessly integrate models which operate in different areas to produce an end to end physical risk analysis, or to experiment with different approaches.

2 Model description

2.1 Overview

A high-level view of the physical risk modelling framework is shown in Figure 1.

Data sets of *hazard indicators*⁵ are obtained from a variety of sources. For modelling acute hazards, datasets typically comprise probability distributions of events such as inundations, periods of drought or periods of high wind. These data sets might, for example, specify the annual probability of occurrence of an event (e.g. wind storm) of a certainty intensity (e.g. maximum gust wind speed) for some specified year in the future. An important class of such data sets are the *hazard maps*,

⁵ a hazard indicator is measure used to quantify a hazard.

described later in this section.

Vulnerability models are applied to different types of asset and different types of hazard. A vulnerability model will use information about an asset, typically its geolocation, type and relevant characteristics, and will source hazard indicator data from one or more *hazard models*. The vulnerability model will construct from these inputs an *impact distribution*, a probability distribution describing the potential impacts on the asset.

In the case of acute hazards, in order to derive the impact distribution vulnerability models will typically construct both:

- Hazard event distributions: probability distributions of events that impact the assets *at their locations*, derived from hazard data sets,
- Vulnerability distributions: conditional probability distributions of the impacts on the assets of events of given intensity.

The vulnerability model uses these quantities to derive distributions of impact for each asset. An impact might be, for example, damage to the asset, expressed as a fraction of the asset value. The financial risk model calculates financial measures from impact distributions.

Within the OS-C modelling framework, models are interchangeable and allow forms of composition. That is, different choices of vulnerability model may be used for a particular asset and a vulnerability model may use different hazard data sets for its calculation. The intention is to allow a risk calculation to be built from an ecosystem of hazard and vulnerability models according to the requirements of the model owner.

2.2 Hazard model

As described above, one or more hazard models are used to provide the hazard indicators required by vulnerability models. Vulnerability models request information from hazard models based on location and spatial extent of the assets.

Hazard indicator data sets are available from a number of sources both publicly-available and proprietary. A recent paper [2] provides a review of sources of the ‘*forward-looking climate-conditioned hazard data sets*’, as the authors describe the data, that are required by the methodology presented in this document. [32] provides technical information on the data sets provided by the Network for Greening the Financial System (NGFS)

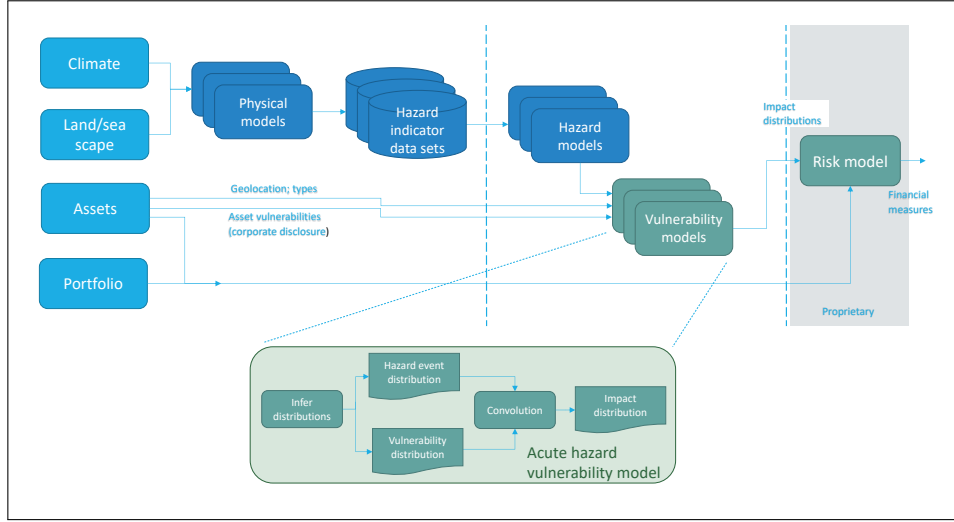


Figure 1: Physical risk model components.

scenarios and provides a useful introduction to scenarios and in particular Representative Concentration Pathways (RCPs) and Shared Socioeconomic Pathways (SSPs) as well as the Coupled Model Intercomparison Project (CMIP) of global climate models data sets. CMIP data sets are very often key inputs into the derivation of forward-looking climate-conditioned hazard indicators.

Subsequent sections include details of both vulnerability models and sources of hazard indicator data sets for each type of hazard.

2.3 Acute vulnerability model

The *vulnerability model* determines how an asset is impacted by a hazard. The impact is a quantity from which financial loss can be inferred, but is not itself a monetary value. For example, an impact might be the damage sustained to a building as a fraction of its value or the annual loss of energy output of a power station as a fraction of its annual output⁶. In each case, a further (financial) model is required to translate the impact to a change in asset value and thereby a change in financial measure. In principle an impact might lead to an increase or decrease in value.

Catastrophe models sometimes define a quantity ‘damage’, and talk about ‘damageability’. ‘Damage’ and ‘impact’ are analogous quantities here but

⁶ A systemic change in annual output changes asset value, since this is partly determined by the expected future cash flows generated by the asset.

‘impact’ is better-suited to situations where there is, say, a decrease in output efficiency of a plant as a result of a period of higher temperatures; that is, the impact can be a damage or a disruption.

Vulnerability models as used in physical risk calculations may overlap with those of catastrophe models. OS-C aims to support a wide range of models, but it is desirable to identify approaches that generalize a large class of these. One such approach is adopted from Oasis LMF [23]. The first assumption behind this is that a model should capture two important types of uncertainty, doing so by representing each by a probability distributions:

1. Uncertainty as to the frequency and intensity (or severity) of events that potentially lead to a change in asset value. This is sometimes called the *primary uncertainty*
2. Uncertainty as to the vulnerability of assets to events (i.e. response of assets to events of a given intensity), the *secondary uncertainty*

These quantities are defined more precisely in 2.3.4. Impact can be modelled using a *mean impact curve* (or *mean damage curve* in catastrophe modelling nomenclature). This is a curve relating an event intensity to an impact (e.g. a wind storm with a given maximum gust speed will cause a given fractional damage to a property). In general, however, there is uncertainty as to the impact on an asset to an event of a given intensity – in the example, the wind may cause mild or severe damage. For this reason, the vulnerability is represented rather as a two dimensional function.

A second assumption is that the probabilities of such events may not be readily represented by distributions such as beta, gamma, beta-Bernoulli or truncated Gaussian and may be complex and multi-modal. Discrete probability distributions are therefore used in order to represent the range of possible distributions: a non-parametric approach. In the mathematical description below, the continuous form of the model is first given followed by a discretisation intended for the capture of non-parametric distributions.

2.3.1 Mathematical description of vulnerability models

The output of the vulnerability model is an impact, d for an asset and for a given year⁷. Physical climate risk analyses are generally concerned with

⁷ d for ‘damage/disruption’ is used to denote impact as i is reserved for indexing.

how impacts change as a result of changes to the climate. In contrast with catastrophe modelling where an impact is typically calculated according to a historical baseline, a climate risk calculation would typically compare the baseline to an impact calculated under a particular climate scenario and period in the future.

The impact of hazards on a portfolio of n assets is a (multivariate) probability distribution as a result of the primary and secondary uncertainties. The impact in a given year on a single asset is a random variable, D , with (marginal) probability density function $f_D(d)$. The probability of an impact exceeding d (the *exceedance probability*), $F'_D(d) = \mathbb{P}[D > d]$ is given by:

$$F'_D(d) = \int_d^\infty f_D(u) du \quad (1)$$

This is related to the cumulative probability $F_D(d) = \mathbb{P}[D \leq d]$ by $F'_D(d) = 1 - F_D(d)$. Exceedance probabilities are a popular measure in catastrophe model and the use of F for cumulative probability and F' for exceedance probability will be used hereafter to avoid confusion.

The value of a hazard indicator, which quantifies the intensity of the phenomenon to which the hazard refers (we also use the term ‘hazard intensity’) is given by random variable S . This has marginal probability density $f_S(s)$. This is the probability density of an event occurring in a given year with intensity s . This distribution captures the primary uncertainty.

We define a conditional probability density function $f_{D|S}(d, s)$ to be the probability that given the occurrence of a hazard indicator s , an impact occurs of size d . This distribution captures the secondary uncertainty.

The impact distribution is then given by:

$$f_D(d) = \int_{-\infty}^\infty f_S(s) f_{D|S}(d, s) ds \quad (2)$$

Note that $f_d(d)$ is identical to the *effective damageability* distribution of Oasis LMF[37] and can be described as the ‘effective impact’. It is a marginal distribution and as such does not capture any correlation between events nor impacts. In the catastrophe models of Oasis LMF, impacts are sampled from this distribution [37], for example samples of fractional damage, which form the basis of a Monte Carlo calculation. This is done in order to apply

insurance policy terms and conditions which can be complex and non-linear.

In the case of n assets, multivariate random variables are defined $\mathbf{D} = (D_1, \dots, D_d)$ and $\mathbf{S} = (S_1, \dots, S_n)$. We henceforth use the single-asset index i to denote marginal distributions: $f_{S_i}(s_i)$.

2.3.2 Return-period-based approach

The *return-period-based approach* – sometimes called, a hazard-map-based approach – makes use of marginal distributions f_{S_i} for each (asset) location i , and not the joint distribution, $f_S(\mathbf{s})$. By definition, the marginals are related to the joint distribution by:

$$f_{S_i}(s_i) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f_S(s_1, s_2, \dots, s_n) ds_1 \dots ds_{i-1} ds_{i+1} \dots ds_n \quad (3)$$

$f_{S_i}(s_i)$ is usually inferred from a type of hazard indicator data set which are known as *hazard maps*.

Hazard maps. Hazard maps are three-dimensional data sets from which intensities of hazard indicators can be looked up for different locations and different return periods, i.e., $h(x, y, \tau)$ provides hazard indicator intensity at location (x, y) for return period τ . That is, h is the hazard indicator intensity such that the average time between events with intensity higher than h is τ .

Remark 2.1. More precisely, if we define X_t^h as the number of events in a given period of length t such that the intensity is greater than h , then, the return period (in years) satisfies [30],

$$\mathbb{E} \left(X_t^{h(x,y,\tau)} \right) = \frac{t}{\tau}. \quad (4)$$

For instance, assuming that X_t^h follows a Poisson distribution with parameter λ^h (see below), that is, $X_t^h \sim \text{Poi}(\lambda_t^h)$, then,

$$\mathbb{E} \left(X_t^{h(x,y,\tau)} \right) = \lambda_t^h,$$

where λ^h is the intensity. Thus, $\lambda_t^h = t/\tau$. By definition, we set $X_1^h := X^h$, that is, if there is no subindex we are assuming one year. $\diamond$

In order to use hazard maps to derive probabilities, we must specify the model of probability of occurrence of events with intensity higher than h

assumed by the data set. Occurrence may be modelled by a Poisson distribution as in Equation 5. This gives the probability of k occurrences in time interval t where τ is the return period.

$$\mathbb{P}[X_t^h = k] = \frac{(t/\tau)^k}{k!} e^{-\frac{t}{\tau}} \quad (5)$$

Remark 2.2. It can be noted that by Taylor's Theorem,

$$\mathbb{P}(X_t^h \geq 1) = 1 - e^{-t/\tau} = t/\tau + o(t/\tau) \approx t/\tau, \quad (6)$$

where the last approximation is valid if t/τ is “small”. $\mathbb{P}(X_t^h \geq 1)$ is also known as the *occurrence exceedance probability*: the probability that the intensity of the highest-intensity event in t exceeds h . Using this notation, F'_S is more precisely defined: $F'_S(h) = \mathbb{P}(X_1^h \geq 1)$. That is, F'_S is the occurrence exceedance probability. $\diamond$

Alternatively, the number of occurrences can be modelled as a Binomial distribution as in Equation 7, which provides the probability that k years have at least one occurrence, for a period of n years, assuming that τ is specified in years. This is given by

$$\binom{n}{k} (1/\tau)^k (1 - 1/\tau)^{n-k}. \quad (7)$$

According to Equation 7, the probability that in a single year there is at least one event with intensity of h or higher is $1/\tau$. Unless otherwise specified, this is the interpretation used for τ . Note that for Equation 5 this relationship only applies approximately, see (6).

F_{S_i} can then be inferred from the hazard map for point-like assets. The curve $h(x_i, y_i, \tau)$ is looked up, providing τ and thereby annual (occurrence) exceedance probabilities for different intensities, h .

Remark 2.3. More precisely, let us define occurrence exceedance probability, $F'_{S,t}(\cdot)$, as

$$F'_{S,t}(s) := \mathbb{P}(\exists S \text{ in the period } t / S > s) = \mathbb{P}(X_t^s \geq 1). \quad (8)$$

How does this relate to the cumulative probability of each individual event, $F_S^{(e)}$? As before, $F'_S := F'_{S,1}$. Then, if we consider a collection of identically distributed variables for each event $\{S^k\}_{k \in \mathbb{N}}$:

$$\begin{aligned} 1 - F'_S(s) &= \sum_{k=0}^{\infty} \mathbb{P}(S^1 \leq s, \dots, S^k \leq s) \mathbb{P}(X^{H=0} = k) \\ &= \sum_{k=0}^{\infty} \mathbb{P}(S \leq s, \dots, S \leq s) \mathbb{P}(X^{H=0} = k). \end{aligned}$$

Note that $X^{H=0}$ represents the number of events in the given period, no matter the intensity. Using Sklar's Theorem, there exist copulas C_k such that

$$1 - F'_S(s) = \sum_{k=0}^{\infty} C_k \left(F_S^{(e)}(s), \dots, F_S^{(e)}(s) \right) \mathbb{P}(X^{H=0} = k) =: G(F_S^{(e)}(s)).$$

By the rectangle inequality for copulas, we know that $\tilde{C}_k(F_S^{(e)}(s)) := C_k(F_S^{(e)}(s), \dots, F_S^{(e)}(s))$ is non-decreasing. If furthermore, it is strictly increasing, it is sufficient⁸ to guarantee that G is increasing. So we can recover $F_S^{(e)} = G^{-1} \circ O$. For instance, if we assume independence and a Poisson process, using the Taylor series of the exponential function,

$$1 - F'_S(s) = \sum_{k=0}^{\infty} F_S^{(e)}(s)^k \frac{\lambda^k}{k!} e^{-\lambda} = e^{F_S^{(e)}(s)\lambda - \lambda}, \quad (9)$$

where $\lambda := \lambda^{H=0}$. Thus, the relation between exceedance probability and occurrence exceedance probability is the following

$$F_S^{(e)}(s) = \frac{1}{\lambda} \log(1 - F'_S(s)) + 1.$$

Note that in this case, by (6),

$$F'_S(s) \approx \frac{1}{\tau_s}.$$

◇

In the case of a point-like asset, the look up is from spatial coordinates (x_i, y_i) . A hazard map will have an associated co-ordinate reference system (CRS). For example the CRS of whole-globe maps is often the WGS84 World Geodetic System (EPSG:4326). In this case (x_i, y_i) represent longitude and latitude under that CRS.

Effective impact. Once F_{S_i} and thereby f_{S_i} is obtained, Equation 2 can be applied to obtain the impact distributions for each location i :

$$f_{D_i}(d_i) = \int_{-\infty}^{\infty} f_{S_i}(s_i) f_{D_i|S_i}(d_i, s_i) ds_i \quad (10)$$

⁸ Consider the case of $\mathbb{P}(N = 2) = 1$ and the countermonotonicity copula $C_2(u, v) = (u + v - 1)_+$, then $G|_{[0,1]}$ is not invertible, so $F_S^{(e)}$ cannot be always recovered from O .

In order to aggregate impacts over a portfolio, the dependency structure must be provided; in general, such a dependency structure is specified by a copula[22]. This may be derived via a heuristic; for example a ‘worst case’ dependency structure could be provided to obtain an upper bound, perhaps assuming 100% correlation between impacts. In general the approach in this section is not intended for cases where accurate treatment of the dependence between the impacts is needed, but rather for:

- analyses of single assets,
- quick calculations intended to give an upper bound of an impact, perhaps to identify areas of particular risk, or
- certain calculations where intra-asset correlation is not required, e.g. expected total ground-up loss.

The Sklar theorem of copula theory states that for n random variables $(D_1, \dots, D_n)$ with joint cumulative density function (CDF) $F_D(d_1, \dots, d_n) = \mathbb{P}[D_1 \leq d_1, \dots, D_n \leq d_n]$, there exists a copula $C : [0, 1]^n \rightarrow [0, 1]$ such that

$$F_D(d_1, \dots, d_n) = C(F_{D_1}(d_1), \dots, F_{D_n}(d_n)). \quad (11)$$

As a reminder, $F_{D_i}(d_i) = \mathbb{P}[D_i \leq d_i]$ is the marginal distribution of random variable D_i , $i = 1, \dots, n$.

In the general case, Monte Carlo approaches can be used to sample from F_D . The approach is:

1. Sample vector $\mathbf{u}$ from C ;
2. Calculate samples for vector $\mathbf{d}$ using the relationship $d_i = F_{D_i}^{-1}(u_i)$.

The samples can then be used to construct a total damage, for example, via $\sum_i d_i$.

Under heuristic approaches a Gaussian copula might be chosen:

$$C_K^{\text{Gaussian}}(\mathbf{u}) = \Phi_n(\Phi^{-1}(u_1), \dots, \Phi^{-1}(u_n); \mathbf{K}), \quad (12)$$

where $\Phi(z)$ is the CDF of a standard normal variable, and $\Phi_n(\mathbf{z}; \mathbf{K})$ denotes a joint standard normal multivariate CDF with mean zero and correlation matrix $\mathbf{K}$. Samples $\mathbf{u}$ can be obtained by the approach:

1. Sample vector $\mathbf{z}$ of correlated normal random numbers,
2. Calculate samples for vector $\mathbf{u}$ using the relationship $u_i = \Phi_i(z_i)$.

As an example, if impact distributions represent damage to buildings as a result of inundation then one may attempt to model damage to two buildings in close proximity as being highly correlated. However catastrophe model practitioners might point out that even such considerations as the presence or absence of kerb stones and availability of sand bags are highly significant so any such assumption is prone to error. If the buildings are far apart (say in different continents) then the correlation is likely to be close to zero however. The two extremes of 0% and 100% correlation of impacts are special cases of this general approach and it may be appropriate to run calculations with such cases to obtain an estimate of the impact of correlation – rather than to attempt to model correlation more precisely.

2.3.3 Event-based approach

As mentioned in the previous section, where an accurate treatment of the dependence between impacts is needed, a heuristic is unlikely to be adequate. This can occur, for example, where it is required to model the occurrence of a 1 in 250 years ‘worst-case’ event, the event being ‘worst-case’ for a specific portfolio⁹. In such cases, it is necessary to model the dependence of hazard intensity and vulnerability explicitly. In an event-based approach, this is achieved by calculating the impact of a large number of events. An event might, for example, be a flood or storm affecting a particular geographical region. Events can be:

- historical events, or
- synthetic, ‘plausible’ events.

In the case of synthetic events, the event may itself be probabilistic; at a particular time, in a particular part of the region, the intensity of the hazard might be represented by a probability distribution. However historical events are typically deterministic and synthetic event sets are often deterministic also; unless otherwise stated, this will be assumed to be the case.

⁹ In general, a ‘1 in 250 year’ event for a portfolio is an ambiguous statement and requires the definition of a measure. For example the ‘1 in 250 year’ *ground-up loss* of a portfolio is unambiguous. For a single asset, ‘1 in 250 year’ is unambiguous if impact is assumed to be a non-decreasing function of hazard intensity.

Relationship to hazard maps. Events may be specified with respect to hazard maps; that is the hazard data comprises both hazard map and event set. Typically in such cases the severity of an event is specified using a return period which can then be related back to the hazard map.

More formally, for each event, index j , a function z_j is defined such that $\tau_{i,j} = z_j(x_i, y_i)$. Here $\tau_{i,j}$ is the severity of the event specified as a return period for asset index i and event index j . (x_i, y_i) are the coordinates of the asset in the relevant CRS.

For the given event, $\tau_{i,j}$ is then used to look up a hazard intensity, from the hazard map:

$$s_{i,j} = H(x_i, y_i, \tau_{i,j}) \quad (13)$$

Note that here $s_{i,j}$ is deterministic *for a given event* and we have simply for the impact distribution for event j :

$$f_{D_{i,j}}(d_i) = f_{D_i|S_i}(d_i, s_{i,j}) \quad (14)$$

For each event and each asset, we can then draw m samples from $f_{D_{i,j}}(d_i)$. The samples across all assets for a single event are drawn jointly using a copula as per Equation 11 to capture the dependence structure of impacts. Such an approach can be important if levels of damage sustained under a given hazard intensity are correlated between assets, which might be the case if the assets are similar in construction. However, by default samples are assumed to be independent. That is the approach is:

For each of n events, and for each of m samples:

1. sample vector $\mathbf{u}$ of uncorrelated random numbers $\mathbf{U} \stackrel{iid}{\sim} U[0, 1]$;
2. calculate vector $\mathbf{d}$ using the relationship $d_i = F_{D_{i,j}}^{-1}(u_i)$.

The resulting $n \times m$ sets of impacts for the portfolio of assets is the inputs into the financial model.

Constant severity regions Under a constant severity region approximation, $\tau_{i,j}$ is a constant across each asset i for a particular event j .

2.3.4 Discrete form of acute vulnerability model

The continuous forms of expressions for the impact distribution are given by Equation 10 and Equation 14, the latter being the special case of deterministic hazard – generally encountered for one specific event. Here a discrete version of Equation 10 is defined.

There are n_h intensity bins with index q such that $q \in \{1, \dots, n_h\}$. $\sigma_q^{(h)}$ is defined to be the probability that a hazard event of type h occurs in a given year with an intensity that falls in bin q . With random variable $S^{(h)}$, now with superscript, being the intensity of hazard type h :

$$\sigma_q^{(h)} = \mathbb{P} \left[s_q^{(h, \text{lower})} < S^{(h)} \leq s_q^{(h, \text{upper})} \right] \quad (15)$$

That is, $s_q^{(h, \text{lower})}$ and $s_q^{(h, \text{upper})}$ define the range of bin q . For the avoidance of doubt, $\sigma_q^{(h)}$ is related to the continuous probability density $f_S(s)$ by:

$$\sigma_q^{(h)} = \int_{s_q^{(h, \text{lower})}}^{s_q^{(h, \text{upper})}} f_S(s) ds \quad (16)$$

We define $v_{pq}^{(h, b)}$ to be the conditional probability that *given* the occurrence of an event associated with a hazard of type h and with intensity $s^{(h)} \in (s_q^{(h, \text{lower})}, s_q^{(h, \text{upper})}]$ there is an impact of type b , $d^{(b)} \in (d_p^{(b, \text{lower})}, d_p^{(b, \text{upper})}]$. b may be, for example, damage incurred expressed as a fraction of the asset present value.

$$v_{pq}^{(h, b)} = \mathbb{P} \left[d_p^{(b, \text{lower})} < D^{(b)} \leq d_p^{(b, \text{upper})} \mid s_q^{(h, \text{lower})} < S^{(h)} \leq s_q^{(h, \text{upper})} \right] \quad (17)$$

The definition of an event type h includes a time interval e.g. h is the occurrence of an inundation in the locale of the asset *within a one year period*. b is, for example, the fractional damage to the asset.

$\delta_p^{(h, b)}$ is defined to be the marginal probability of impact $D^{(h, b)}$ in the range $d_p^{(b, \text{lower})} < D^{(h, b)} \leq d_p^{(b, \text{upper})}$ occurring as a result of an event of type h .

$$\delta_p^{(h, b)} = \mathbb{P} \left[d_p^{(b, \text{lower})} < D^{(h, b)} \leq d_p^{(b, \text{upper})} \right] \quad (18)$$

From the definition of conditional probability:

$$\delta_p^{(b)} = \sum_q v_{pq}^{(h,b)} \sigma_q^{(h)} \quad (19)$$

If only the mean impact curve is available, then it is possible to create the matrix such that $v_{pq} \in \{0, 1\}$. The matrix then provides a simple mapping from intensity to impact; if the number of intensity and response bins is equal then matrix $\mathbf{v}$ is simply the identity matrix. However, note that these simplifications exclude from the model any uncertainty in the parameters¹⁰.

Multiple occurrence of events. Note that $\sigma_q^{(h)}$ is the probability of occurrence of *at least one event* with intensity in bin q in a year and the vulnerability, v_{pq} gives the probability of impact given at least one event has occurred. Some care must therefore be taken when using probabilities v_{pq} calibrated from single events as there is an implied approximation that either probability of multiple events is small and/or that impact is well-modelled as a single impact from the most intense event for a given year.

2.3.5 Importance of secondary uncertainty

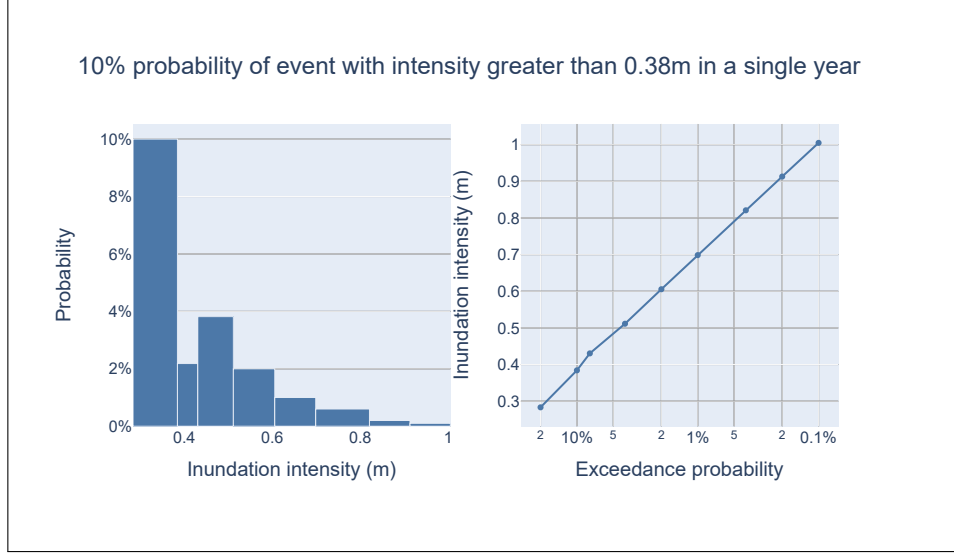
The importance of the vulnerability matrix as opposed to mean damage curve (or vector) is emphasized above; see also [36] for a discussion of this point. This is true not only in cases where the underlying distribution of an impact, for example a fractional damage, can be inferred from empirical data; see for example Figure 3. This is arguably *more* important where data is limited in order that approximate data can be incorporated into the model in a way that the impact of the approximations can be well-understood.

Vulnerability data may be provided by

- modelling of asset vulnerability based on asset characteristics and/or historical data, or
- 'calibrated' vulnerabilities, for example based on realized insurance claims

Physical risk models may make use of so-called 'bulk assessment' approaches for certain assets, where precise vulnerability information is not available and

¹⁰ A better approach would be to estimate the standard deviation of the distributions from which the mean impact curve was calculated and to incorporate this.



The exceedance curve of event intensity at the asset location is shown on the right. The event intensity in this example is inundation depth in metres. Exceedance is a cumulative probability. As an example, the probability of an inundation event occurring within a single year of intensity 0.91m or greater is 0.2%. An exceedance probability is the reciprocal of the return period; it could equivalently be said that the 0.91m intensity event occurs with a return period of 500 years. The exceedance curve can be converted to a histogram of probabilities. Here the n_h bins have ranges $(s_q^{(h,lower)}, s_q^{(h,upper)}]$. For example, the first bin has range (0.28m, 0.38m]. The second bin has range (0.38m, 0.43m]; that is $s_2^{(h,lower)} = 0.38\text{m}$ and $s_2^{(h,upper)} = 0.43\text{m}$. $\sigma_2^{(h)} = 0.02$.

Figure 2: Event intensity exceedance curve (right) and corresponding histogram (left).

less precise estimates of the damage/disruption of the asset are used. The presence of such estimates in an overall model may, or may not, materially impact the accuracy of the results, but it is important that this impact can be assessed. By quantifying the uncertainty in the response estimates, a distribution of financial losses is ultimately obtained from which the model user can derive the impact of the approximation.

Including epistemic uncertainty in vulnerability. In forms of bulk-assessment, and indeed in other cases, a common occurrence is that insufficient information exists with which to characterize an asset. This is an example of an epistemic, as opposed to aleatory, uncertainty. The epistemic uncertainty, and its impact, can be included in the model in the following way.

We extend Equation 17, by including a new discrete random variable, A ,

which is an integer index, $A \in \{0, \dots, n_a\}$, indicating the type of the asset. We define $v_{pqa}^{(h,b)}$ to be the probability that $D^{(h,b)} \in (d_p^{(b,lower)}, d_p^{(b,upper)}]$ given that $S^{(h)} \in (s_q^{(h,lower)}, s_q^{(h,upper)}]$ and the asset in question is of type a :

$$v_{pqa}^{(h,b)} = \mathbb{P} \left[D^{(h,b)} \in (d_p^{(b,lower)}, d_p^{(b,upper)}] | S^{(h)} \in (s_q^{(h,lower)}, s_q^{(h,upper)}], A = a \right] \quad (20)$$

The conditional probabilities can then be combined:

$$v_{pq}^{(h,b)} = \sum_a v_{pqa}^{(h,b)} \mathbb{P}[A = a] \quad (21)$$

$\mathbb{P}[A = a]$ is the *prior* probability that the asset is of type a . This may be obtained from knowledge of the make-up of a portfolio.

Note that through the application of Equation 19 the impact distribution now depends on the prior probabilities. To illustrate why it is reasonable that this should be the case, say we have two types of assets in our portfolio. Type A is vulnerable to a hazard with an intensity that has a relatively short return period of 50 years, whereas type B is invulnerable to all hazards but those with a vanishing small probability of occurrence. To derive the probability that an asset is damaged by a certain amount in a given year using Equation 21, we must allow for the possibility that the asset may be of type A and may therefore be damaged as a result of 50 year events.

Epistemic uncertainty as source of error. An alternate approach is to treat epistemic uncertainty as a source of error rather than, or in addition to, including it in the vulnerability as we have done here. This might be driven by the observation that as information as to the identity of assets is improved then the exceedance probability of a certain impact will change. This can be achieved by running an ensemble of calculations, changing the prior probabilities in each case.

2.3.6 Interpolation of probability distributions

Cases arise where the event distributions and vulnerability distributions are not defined for a common set of intensity bins and interpolation is therefore required. The question then arises of how probability density is distributed within bins. The choice is model-specific and customizable, but here two common cases are described.

- Probability density constant across bin: linear interpolation of cumulative probability function
- Probability density changes linearly across bin: quadratic interpolation of cumulative probability function

[Add equations and example plots here]

Hazard data sets might also contain instances of ‘point-probabilities’, for example where there is a finite probability that the intensity of an event takes a single value. These represent Dirac delta functions in the probability distribution, steps in the cumulative probability function. There is the option of retaining these as delta functions (bins of zero width), but in some cases it may be necessary to make assumptions about how these the probability might be distributed across a bin.

[Add equations and plot of step-CDF with interpolation; exemplify by ‘damage threshold’]

2.3.7 Probability bins from hazard maps

From Equation 16 the probability of an event occurring with hazard intensity in bin q is expressed in terms of the probability density f_S (dropping superscript h for clarity):

$$\sigma_q = \int_{s_q^{(\text{lower})}}^{s_q^{(\text{upper})}} f_S(u) du = \int_{s_q^{(\text{lower})}}^{\infty} f_S(u) du - \int_{s_q^{(\text{upper})}}^{\infty} f_S(u) du \quad (22)$$

The exceedance probability F'_S is defined as:

$$F'_S(s) = \int_s^{\infty} f_S(u) du \quad (23)$$

from which we can write:

$$\sigma_q = F'_S(s_q^{(\text{lower})}) - F'_S(s_q^{(\text{upper})}) \quad (24)$$

Using Equation 24, a set of probability bins for the hazard event can be inferred from an exceedance probability curve. An exceedance probability curve can readily be inferred from a return-period curve using the result

that the annual exceedance probability is the reciprocal of the return period expressed in years.

As an example, suppose that we have a hazard map for flood which contains return periods of 2, 5, 10, 25, 50, 100, 250, 500 and 1000 years. For a certain latitude/longitude the flood depths corresponding to the 9 return periods are, in metres: 0.06, 0.33, 0.51, 0.72, 0.86, 1.00, 1.15, 1.16 and 1.16. The data is shown together with the exceedance probability in Table 1.

Table 1: Example hazard event data. The exceedance probability calculated via $1/\tau$ is provided as well as the value assuming a Poisson distribution. There is a significant difference for short return periods only.

Return period (years)	Flood depth (m)	$F'_S(s)$	Poisson $F'_S(s)$
2	0.06	0.5	0.39347
5	0.33	0.2	0.18127
10	0.51	0.1	0.09516
25	0.72	0.04	0.03921
50	0.86	0.02	0.01980
100	1.00	0.01	0.00995
250	1.15	0.004	0.00399
500	1.16	0.002	0.00199
1000	1.16	0.001	0.00099

The flood depths become the bin edges of the probability distribution and the probabilities are calculated from Equation 24.

Remark 2.4. As before, F'_S is the *occurrence* exceedance probability. Similarly to Remark 2.3, for $u > l$,

$$\begin{aligned}\mathbb{P}(\exists S / S \in (l, u]) &= \mathbb{P}(\exists S / S > l) - \mathbb{P}(\exists S / S > u) \\ &= \mathbb{P}(X^l \geq 1) - \mathbb{P}(X^u \geq 1) = F'_S(l) - F'_S(u),\end{aligned}$$

◇

For example, the probability of occurrence of a flood with depth in the range (0.86m, 1.00m] is $0.02 - 0.01 = 0.01$ ¹¹. Note that in defining a set of bins in this way, no assumption about the interpolation between the flood depths is required. However, if we assume this to be linear then this implies that the probability density is constant across each bin since $f_S = \frac{dF'_S(s)}{ds}$.

¹¹ Care is needed at either end of the curve. There is a 0.001 probability that flood depth exceeds 1.16m in this example; should this be included in the (point-like) 1.16m bin?

2.3.8 Vulnerability distributions and heuristics

For some assets, a vulnerability matrix may be available, corresponding to the ‘ideal’ case of Figure 3. That is, for each intensity value a probability distribution of the impact (damage/disruption) is given. In other cases, only the impact itself is available for a given hazard intensity – a damage curve – together with some measure of uncertainty. Here the probability distribution is unknown, but it is at least possible to fit some choice of distribution to the descriptive statistics (e.g. mean and standard deviation).

Given that the distribution of fractional impact – e.g. damage as a fraction of asset value or disruption as a fraction of output capacity – is in the range (0, 1), heuristic choices of distributions include Beta and Truncated Gaussian [21]. It should be emphasized that neither distribution is likely to be correct, especially in lacking multi-modality and fat-tails, but provide a method by which uncertainty in the impact can be taken into account.

Modelling impact using a Beta distribution. The cumulative probability function of a Beta distribution is given by:

$$F_{\text{Beta}}(x) = \frac{B(x; a, b)}{B(1; a, b)}, \quad (25)$$

where $0 < x < 1$ and $B(x; a, b)$ is the incomplete Beta function:

$$B(x; a, b) = \int_0^x t^{a-1} (1-t)^{b-1} dt. \quad (26)$$

If the mean, μ and standard deviation, σ of the impact distribution are known then [21]:

$$a = \frac{(1-\mu)}{c^2} - \mu, \quad (27)$$

$$b = \frac{a(1-\mu)}{\mu} \quad (28)$$

where

$$c = \frac{\sigma}{\mu} \quad (29)$$

In order to calculate the impact, the set of bins p that define the impact probabilities δ_p of interest are first defined. The vulnerability matrix v_{pq} of Equation 17 is then calculated. In order to apply Equation 25, a and b are calculated using mean and standard deviations of impact calculated at the mid-point of each intensity bin. We then have:

$$v_{pq} = F_{\text{Beta}}(d_p^{(\text{upper})}; a_q, b_q) - F_{\text{Beta}}(d_p^{(\text{lower})}; a_q, b_q) \quad (30)$$

a_q and b_q are the values of a and b calculated from means and standard deviations calculated at the intensity bin centre $s_q = \frac{s_q^{(\text{lower})} + s_q^{(\text{upper})}}{2}$.

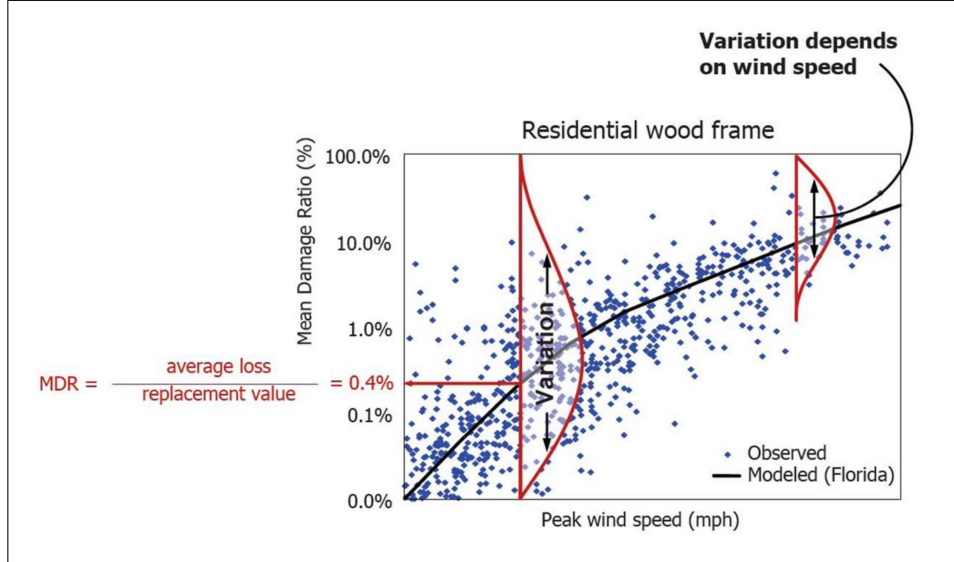


Figure 3: Taken from Lagacé (2008) Catastrophe Modeling, Université Laval. Mean damage curve as an approximation to an underlying set of distributions, modelled using a vulnerability matrix.

2.4 Chronic vulnerability models

2.4.1 Threshold-based vulnerability models

It is possible to construct a framework for implementing a variety of chronic vulnerability models, including those defined in [14]. In contrast to acute model hazard indicators – which comprise quantities such as flood depth or wind speed for different return periods – this class of chronic hazard indicators are based on thresholds. Examples are:

- number of days in a year with near-surface maximum air temperature above temperature τ where $\tau \in \{25^\circ\text{C}, 30^\circ\text{C}, 35^\circ\text{C}, \dots 60^\circ\text{C}\}$.
- number of months in a year with SPEI [39] below τ where $\tau \in \{-1, -1.5, -2, -3, -3.5, -4\}$.
- number of annual degree days above τ where $\tau \in \{18^\circ\text{C}, 20^\circ\text{C}, 22^\circ\text{C}\}$.

Note that the indicators are informative in their own right, but also describe a probability distribution for an underlying quantity. The indicators are based on certain thresholds and a year fraction for which indicator value is above or below the threshold.

Similar to the acute case, σ_q is defined to be the fraction of time (which can also be interpreted as a probability) that the chronic indicator, S , e.g. daily average near surface air temperature or SPEI index, is within a given range.

$$\sigma_q = \mathbb{P} \left[s_q^{(\text{lower})} < S \leq s_q^{(\text{upper})} \right] \quad (31)$$

The disruption function $d(s)$ gives the fractional decrease in the rate of revenue generation for the asset. For example a value of 10% means that the asset is generating revenue at 90% of its normal rate. A loss distribution is then defined; this gives the probability distribution of occurrence of reduced production:

$$\begin{aligned} \delta_q &= \mathbb{P} \left[d_q^{(\text{lower})} < D \leq d_q^{(\text{upper})} \right] \\ d_q^{(\text{lower})} &= d(s_q^{(\text{lower})}) \\ d_q^{(\text{upper})} &= d(s_q^{(\text{upper})}) \end{aligned}$$

δ_q is not the probability of occurrence of an event but rather the probability of occurrence of reduced production in the given range for any time in a given year. To give an example, S is the average web bulb globe temperature and D is the loss of worker productivity, for the fraction of work done outside with medium intensity. Say that between 35°C and 40°C the loss of worker productivity is on average 95% and there are on average 0.2 days per year where this is the case. On average a fraction $0.2 / 365$ of the year has those conditions and for this fraction productivity is at 5% of its nominal level.

Note the difference with the acute case: $0.2 / 365$ is not the probability of having a year with productivity at 5% of its nominal level. Chronic loss over the year is calculated:

$$\text{Chronic loss} = \sum_q \delta_q \left(d_q^{(\text{lower})} + d_q^{(\text{upper})} \right) / 2 \quad (32)$$

2.5 Aggregation of impacts

For impacts of the same type, b , arising from different events, it is assumed that the impacts are additive, up to a ceiling value¹². If the annual impacts from events with index 1 and 2 are represented by random variables, $Y^{(1,b)}$, $Y^{(2,b)}$ then $Y^{(\text{tot},b)} = Y^{(1,b)} + Y^{(2,b)}$.

If the random variables are uncorrelated, then the aggregated effective impact distribution is given by the convolution:

$$y^{(\text{tot},b)}(r) = \int_{-\infty}^{\infty} y^{(1,b)}(t) y^{(2,b)}(r-t) dt \quad (33)$$

[Add version with discrete binned data.]

2.6 Hazard indicators spatial interpolation

Two dimensional geospatial raster data is typically either ‘pixel-is-area’ or ‘pixel-is-point’. As the name suggests, for pixel-is-area data the pixel value is thought of as filling an area. For pixel-is-point the values are a grid of points. Hazard indicator data sets are generally pixel-is-area. In this case, a raster grid space is defined with i as the spatial x and j as the spatial y coordinate. A point in the space is given by (i, j) with i increasing to the right and j increasing downwards. The top left of the image is at point $(0, 0)$ and the bottom right corner of the top left pixel is at point $(1, 1)$. The bottom right of the image is at (n, m) . The centre of the top left pixel is at point $(0.5, 0.5)$.

The usual convention for coordinates (e.g. Python’s *xarray*) is that the positions of the *centres* of the pixels are provided. For example the spatial

¹² this approximation is only strictly valid for sufficiently small impacts; consider the contrived example of 0.8 fractional damage that occurs from both flood and high wind in the same year.

x coordinates are the transformed values of points in raster space with $i \in \{0.5, 1.5, \dots, n - 0.5\}$.

For pixel-is-area data, ‘floor’ interpolation typically means that the value of the pixel containing the point is returned. For example, the top left pixel value is returned for any point such that $0 \leq i < 1$ and $0 \leq j < 1$. For ‘bilinear’ interpolation the pixel value is taken to be located at the centre of each pixel. This is shown in Figure 4. Note that the use of pixel centres makes the algorithm for bilinear interpolation different to that of pixel-is-point data.

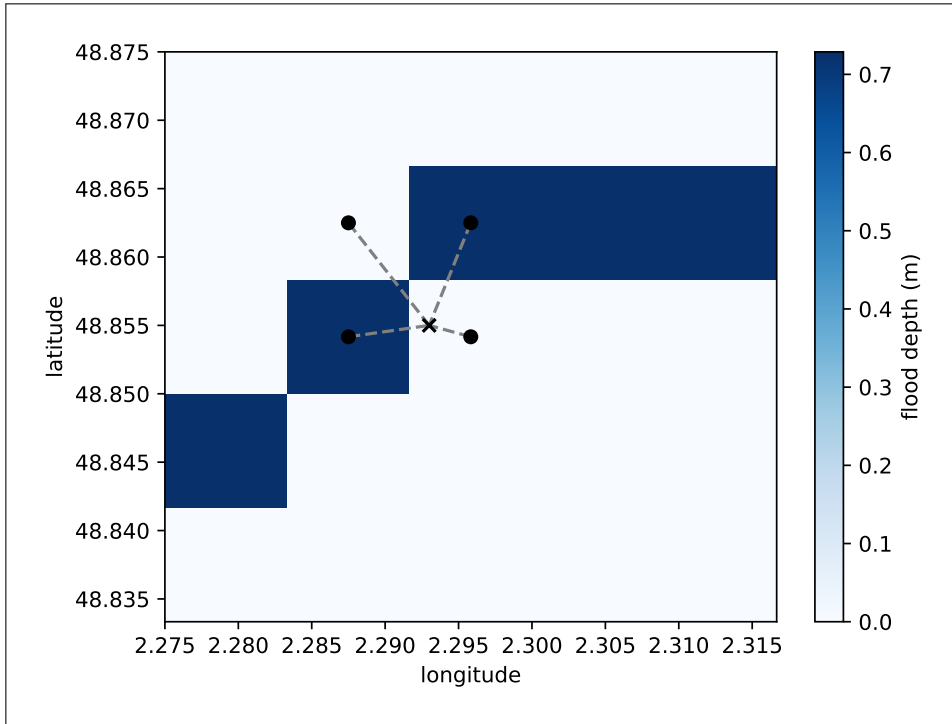


Figure 4: Points used for interpolation under the pixel-is-area convention. The cross marks the asset location in question and the squares are the centres of the four pixels that contribute to the hazard indicator value if using bilinear interpolation.

2.7 Financial model

2.7.1 Transmission channels of risk

Vulnerability models were described in 2.3 as a means to calculate for a portfolio of assets, the damage, disruption to revenue generation or impact on costs, from different hazards. The damage is given as a fraction of the

total insurable value (TIV) of the asset. The disruption is given as a fraction of the revenue generation that is attributable to the asset. How the resulting probability distributions of impact are then used in a financial model depends on the intent of the analysis. Before describing the financial model, it is helpful to give an overview of the different use cases for physical risk models and the different financial products that these can be applied to.

Considering physical risks to lenders such as Banks, a number of activities, corresponding to different financial products can be identified:

- Residential real estate
- Commercial real estate
- Project finance
- Small and medium-sized business / business banking
- Large corporates
- Investment management / own operations
- Financial institutions, sovereigns, and trading book

For each of these, different risk analyses may be needed, representing different use-cases. These use-cases can be split into: *exposure-based*, for example credit granting process/origination and collateral (re-)evaluation; *portfolio-based*, for example portfolio/sectoral analyses, definition/monitoring of risk appetite, Pillar III reporting; and *scenario-based*, for example regulatory stress testing, ICAAP and other scenario analyses.

The modelling approach taken for each use case may differ; exposure-based analyses may take into account the specific characteristics of an asset and its vulnerability to different hazards, whereas portfolio-based analyses may take a more aggregated approach.

For each activity, a financial model is based on the transmission channel of risk:

- Devaluation of collateral,
 - market driven: devaluation of collateral, giving rise to higher loss given default (LGD), as a result of a climate-related change in the market value of (undamaged) assets, or

- from asset impairment: devaluation of collateral, as a result of damage sustained to the asset and/or degradation of its surroundings.
- Deterioration of the client financial strength, for
 - companies: deterioration of financial strength from impact on cashflows resulting in a change in the client probability of default (PD), or
 - individuals: deterioration of financial strength from an individuals income and outgoings.
- Devaluation of an owned asset,
 - market driven: devaluation, causing direct loss, as a result of climate-related change in the market value of (undamaged) assets, or
 - from asset impairment: devaluation, causing direct loss, as a result of damage sustained to the asset and/or degradation of its surroundings.

We take two examples, which are illustrative of a wide range of cases:

Commercial real estate lending. Devaluation of the loan collateral, which is to say the commercial real estate property, market driven, can occur if investors reassess the valuation of exposed and vulnerable properties on the basis of increase in damage-related costs, and/or lower utility. For insured assets, this reassessment may include risk of insurance becoming unaffordable or unavailable. This revaluation may occur in the absence of any damage to the asset itself, although can be associated with an event (e.g. increased incidence of wildfires in the vicinity).

Small and medium-sized business lending. Client financial strength can deteriorate from impact on cashflows related to the restoration of damaged assets, loss of revenue, or increase in costs. Loss of revenue or increase in costs can be from direct impacts (scope A), from withdrawal of infrastructure services (scope B) or from impact on the value chain (scope C). For example, a restaurant may be damaged by a flood and require capital expenditure to restore the asset. The restaurant may also lose revenue during the restoration period or, for example, if the local road network is damaged. Finally, the restaurant may be affected by impacts on its supply chain or changes in customer behaviour.

Scenario-based analyses, of which stress tests are an example, make use of climate-related shocks in order to assess financial impact. Stress tests may make use of narrative scenarios or may take the approach of simulating large number of extreme but plausible events in order to identify a worst-case which is *a priori* unknown. Indeed the two approaches may be combined, assessing the impact of a severe acute event coupled with other aggravating factors.

2.7.2 Climate risk measures and importance of change.

Climate risk models generally differ from catastrophe models in more than their use of climate-conditioned and projected hazard models. In particular, physical climate risk arises from *changes* in climate.

One important question for financial institutions is how changes in climate can affect the credit risk of market participants. Two main channels by which this can occur:

1. changes in the value of assets that are collateral for loans;
2. changes in the probability of damage to assets or disruption to revenue generation of institutions' counterparties which can lead to changes in financial health.

The first of these is likely to manifest in a change of loss given default (LGD) and the second in a change of probability of default (PD).

We take as an example of the first case a real estate asset that is collateral for a loan. The hypothetical asset is exposed to both topical cyclone and flood hazards today, however it is covered by an insurance policy and its current valuation takes into account both historical occurrence of hazards and the policy details. Risk managers concerned with climate risk are likely to be interested in potential changes in value of the asset over the full term of the loan:

1. if acute events become more frequent and severe, assets may become more expensive to insure and less desirable *[Citation]*¹³, which can decrease the asset value;
2. if an extremely severe and widespread event occurs, default may lead to a distressed sale and/or the full insurance payout may not be made.

¹³ For example, to a resident of a house subject to regular flooding.

For operational monitoring, it is therefore likely to be the change in frequency of events and the change in probability of extremely severe events that is particularly of interest; these are the risk drivers.

Definition of measures. For a portfolio, distributions of annual impacts, typically damage and disruption, can be aggregated and converted into distributions of change in portfolio value in a given reporting currency. Measures describing the loss of value are:

1. Annual Exceedance Probability (AEP): the probability that in a given year the aggregated losses of a portfolio will exceed a certain value.
2. Average Annual Loss (AAL): the mean value of the aggregated losses of a portfolio in a given year.

From the argument above, a change in AEP for moderate and severe loss cases can be a useful indicator of risk. The change is from historical baseline to a climate-conditioned projection.

2.7.3 Incorporating physical risk into models of credit risk

This section gives a brief introduction to the incorporation of physical climate risk into credit risk models. Credit models come in a variety of forms and financial institutions often have pre-existing models; the aim of this section is therefore to identify the requirements of physical risk models that allow climate risk factors to be incorporated rather than to define an end-to-end approach.

Taking the first case of changes in the value of assets that are collateral for loans, it is necessary to model how climate change can affect the value of assets and hence LGD. This can be done in several ways. One can take a *hedonic pricing* approach, and model the changes in value from external factors such as likelihood of a flood. Alternatively one can attempt to set a fair price for the additional hazard exposure by estimating the cost of mitigation, either adapting a property or insuring it. It is worth noting that in either approach significant assumptions are required. It may be argued that such calculations are therefore more appropriate for stress tests or to give a bounding calculation intended to estimate provisions, rather than as an adjustment to LGD in existing credit models.

[Add hedonic pricing refs, e.g. Kousky]

In the second case, the focus is on how climate change impacts the financial health of counterparties. Impacts can take the form of damage to assets. Restoring assets to their original condition may require capital expenditure, depending on mitigating factors such as the presence of insurance. Asset restoration may also lead to downtime and loss of revenue. Other impacts of climate change affect revenue generation directly, for example heat-related loss of productivity. The net effect of the impacts can be an increase in debt and decrease in equity of a counterparty with potential impact on PD.

The aggregate amount required to restore assets and aggregate loss of revenue are therefore likely to be important. Integrating these into credit models can be done in several ways. Some types of credit model are based on sets of financial ratios, e.g. debt to equity; here it might be appropriate to shock the ratios based on the physical risk model outputs. *Refs for financial-ratio-based approaches*. A class of credit risk models called ‘structural models’ are also of interest. For medium and large cap firms, a credit default event typically occurs when a firm is not able to meet its debt servicing obligations and it may be assumed that a default event occurs for a firm when its assets are sufficiently low compared to its liabilities. A number of different structural models exist which make various assumptions about how a firm’s assets change over time, how its capital is structured and the nature of its debt.

The earliest structural model was described by Merton in 1974 [20] based on an extremely simple debt structure. Black and Cox [5] introduced an important refinement to the Merton model in 1976. Practical implementations were subsequently created as a result of this foundational work. A notable one of these is the ‘KMV’ model, named after Kealhofer, McQuown and Vasiek, now owned by Moody’s Investors Service, Inc. Such credit models, provide a mathematical framework for incorporating of physical risk [12] and are a likely candidate for an illustrative end-to-end approach.

2.8 Uncertainties in the calculation

2.9 Model limitations

1. Spatial correlation of events: to what extent possible without MC calculation; to what extent is provided / can be inferred from data sets
2. Correlation of vulnerability
3. Data availability

2.9.1 Data availability

Issues related to data availability and relevance are still one of the main limitations of physical risks assessments. If past and future climate data are becoming increasingly available through open-sources portals and tools (e.g. Copernicus, WRI Aqueduct), their availability and their reliability varies widely according to the climate hazard of interest, the region and the modelling process. If the availability of climate data is improving, open-source, asset-level information (required to estimate the exposure of an asset to a give climate hazard) is still seldom available. Such data include the location of assets, their link with owning companies and more generally any damages records that could be used to quantify the response of an asset (or of a type of asset) to a given climate event. Newly-published datasets have been recently released for some sectors but their exhaustiveness remains to be verified. Moreover, many industrial sectors are not covered, thus limiting the application of physical risks methodologies to a diversified portfolio.

Finally, building and applying the correlation between hazard and damage (or impact), as described in [2.3.3](#), requires common distribution between historical events, historical damages and future climate events. In a changing climate, assets and activities will be impacted by more intense events that will not have been experienced either in a given region of the world or even on the whole globe, leading to a potentially large mismatch between historical and future distributions of events.

3 Hazard and vulnerability models

3.1 Inundation

3.1.1 Hazard models

Inundation is modelled as an acute risk using the approach of Section [2.3.4](#). Hazard models compatible with this method provide inundation depths for different annual probabilities of occurrence – or equivalently return periods. The need for sufficient granularity in the set of return periods is discussed in [\[40\]](#).

Inundation comprises three types:

- Coastal

- Pluvial
- Riverine (aka Fluvial)

A number of hazard models are available, both public-domain and commercial (see [physrisk library documentation](#)). The World Resource Institute (WRI) Aqueduct flood model [42] is a notable public-domain global flood model, comprising coastal and riverine inundation, which has relatively high return-period granularity. The data set is based on the modelling approach of [41]. The WRI data set has a spatial resolution of approximately 1 km at the equator.

Higher resolution public-domain sets are available, albeit the ones we are aware of for Europe – and have on-boarded into OS-Climate – cover limited geographical regions. TU Delft provide a pan-European set [25] with a 100 m resolution, covering riverine and coastal hazards. The European Commission Joint Research Centre data catalogue includes a 90 m resolution map, but is restricted to current climate [9].

Commercial flood models provide global coastal, pluvial and riverine hazard maps and event sets with resolution sufficient to model street-level inundation (approx. 5 m resolution or finer). OS-Climate provides connectors for certain model-providers, to facilitate integration of commercial alongside public-domain models. Examples of specialist flood providers who maintain APIs are [JBA Risk Management](#) and [Fathom](#).

3.1.2 Vulnerability models

Notable sources of vulnerability curves are:

- FEMA Hazus [10], who kindly provided an extract of their curve set (see [physrisk library documentation](#)) for access and use.
- A European Commission Joint Research Centre (JRC) model [11].
- (Commercial) The Multi Coloured Manual [28], Middlesex University.

Thermal power generation projects When flood water inundated a thermal power plant, operations are disrupted due to infrastructure damage and/or mechanical failure. The flood module described in [14, 15] is implemented in the *physrisk* library. The inputs are:

- The highest flood water depth per return period due to either coastal or riverine inundation in a buffer zone with a radius of 0.01 arc degrees around the geolocation of the generation unit. It allows to derive:
 - a discrete exceedance probability curve for flood water depth in a climate-conditioned projection;
 - a designed flood protection level defined as the flood water depth interpolated from a baseline for a return period equal to either 10000 years for nuclear plants and 250 years otherwise. Setting the baseline to the historical scenario did not provide satisfactory results. Hence, we rather opt for the year 2030 associated with the selected climate-conditioned projection.
- The disruption curve estimating the number of inoperable days in a year from flood water depth provided that it exceeds the designed flood protection level. It depends on the power production technology, more precisely, the combination of generation technology (gas or steam) and cooling system (dry, recirculating or once-through). The maximum number of inoperable days interpolated for a given flood water depth across potential disruption curves is chosen when turbine and/or cooling system is not provided as an asset archetype.

The output is a discrete disruption probability curve due to the assessed flood risk, that is, either riverine or coastal inundation. It provides the probability in a future projection that the operations of the generation unit are disrupted for a given period expressed as a year fraction.

3.2 Drought

3.2.1 Hazard models

Drought, a significant global natural hazard, leads to substantial economic, environmental, and human distress. Numerous indices have been developed to measure drought intensity. In this framework, standard precipitation evapotranspiration index (SPEI) is used to assess meteorological drought. SPEI is a multi-scalar index designed to take into account both precipitation and potential evapotranspiration as detailed in [39, 3]. Based on this index, drought is modelled as a chronic risk using the number of months per year that the Standardised Precipitation-Evotranspiration Index (SPEI) calculated on a specific timescale (3 or 12 months) is below a given threshold. The SPEI is computed using temprature and percipitation

from General Circulation Models under various Representative Concentration Pathways (e.g., RCP 8.5). Potential evapotranspiration (PET) is estimated from surface daily temperature according to [35] using xclim: Climate services library. For the scope of this methodology, 12-month Standardized Precipitation-Evapotranspiration Index (SPEI) has been considered following [16, 1, 26]. The SPEI at each grid point is calculated for calibration period of (1985 - 2015) and projected on (2015 - 2100) period using xclim: Climate services library.

In literature, SPEI below -1.5 is often considered as severe and extreme drought [33], in this framework, the drought index is measured as number of months per year where the 12-month rolling average of SPEI is below set of thresholds ranging from -3.6 to 0 following [14]. Eventually, the drought index at each grid point is calculated for periods of 2005 (1995-2015), 2030 (2020-2040), 2040 (2030-2050), 2050 (2040-2060), and 2080 (2070-2090) averaged over 20 years.

3.2.2 Vulnerability models

Thermal power generation projects Water levels in rivers and/or reservoirs are low during droughts. Wet-cooled plants might lose access to cooling water or not have sufficient flow rates for cooling effect. The drought module described in [14, 15] is implemented in the *physrisk* library. It excludes temperature effects which are taken into account in a separate module. The inputs are:

- The number of months per year that the 12-month rolling average of SPEI at the geolocation of the generation unit is below a discrete set of thresholds ranging from -3.6 to 0 . An alternative hazard indicator provides only the number of months per year that the 3-month rolling average of SPEI is below -2 . When relying on this alternative hazard indicator, we assume that the number of months per year that the 3-month rolling average of SPEI is below a threshold x lower than -2 satisfies

$$(\text{estimated \#months/year with 3-month SPEI} < -2) \left(\frac{\Phi_{\mathcal{N}(0,1)}(x)}{\Phi_{\mathcal{N}(0,1)}(-2)} \right)$$

given the cumulative distribution function $\Phi_{\mathcal{N}(0,1)}$ of a standard normal random variable $\mathcal{N}(0, 1)$.

- The SPEI-disruption curve estimating the fraction of inoperable capacity with respect to the 12-month rolling average of SPEI. It

applies to once-through or recirculating steam turbine. The largest inoperable capacity ratio interpolated for a given SPEI threshold across potential disruption curves is chosen when turbine and/or cooling system is not provided as an asset archetype.

The output is a discrete disruption probability curve due to drought risks. It provides the probability in a future projection that the operations of the generation unit are disrupted for a given period expressed as a year fraction.

3.3 Heat

Heat is classified as both a chronic and an acute hazard. For example, increased average temperature in a particular area can lower average productivity from labour or make the area less desirable as a place to live, lowering real estate prices. We classify these as risks from chronic hazards. Heat waves are examples of acute hazard events; a period of particularly high temperature might lead to the complete suspension of industrial activity.

Multiple indexes for quantifying heat hazards have been suggested and multiple approaches for the modelling of acute events are present in the literature, e.g. [18]. Similarly, various methods for modelling the vulnerability to heat hazards have been suggested. Analyses of heat wave events are commonly based on Global and Regional Circulation Model (GCM and RCM) outputs [8]. In [6] and [7] the authors analyse ensembles of CMIP6 simulations with and without anthropogenic forcings in order to determine if extreme heat events are attributable to (anthropogenic) climate change. Such attribution analysis is based in part on finding return periods of events (see also [34]). This estimation of return periods for events is directly applicable to acute hazard models.

In order to support a wide range of hazard and vulnerability models, *physrisk* includes the derivation of heat statistics from CMIP6 data¹⁴.

3.3.1 Hazard Models

Chronic hazard models As mentioned above, hazard models and vulnerability models are closely coupled. [44] describes the ‘GZN’

¹⁴ This is somewhat in contrast to the use of the Aqueduct model of [42] for modelling inundation where the complete hazard model is used as-is within *physrisk* – albeit reformatted to handle efficiently the access patterns needed for physical risk calculations.

(Graff-Zivin and Neidell) and ‘WBGT’ (Wet-Bulb Globe Temperature methodology) methods. The statistics required for these methods are derived using bias-corrected and down-scaled data sets. There are multiple sources of data suitable for the estimation of the required statistics, notably the NEX-GDDP-NASA set [38].

Many OS-Climate-generated indicators for chronic heat count the number of days (or weeks) per year over a 20-year period for which a daily (or weekly) average temperature is above a threshold specified in °C. The average temperatures are sourced from down-scaled CMIP data sets. These chronic heat indicators are available in the *phyrisk* library at a predefined set of thresholds for near-surface temperature, WBGT index and water temperature. This allows to infer a discrete probability distribution under some climate-conditioned projections as well as a historical baseline.

Acute hazard models Acute hazard modelling approaches are based on calculating return periods of events in a way analogous to acute inundation models. The calculation of return periods from data sets presents a statistical challenge, dealt with for example by [19].

3.3.2 Vulnerability models

3.3.3 Thermal power generation projects

3.3.3.1 Air temperature

Air temperature primarily affects the generation efficiency of gas turbines and cooling efficiency of air-cooling systems. The air temperature module described in [14, 15] is implemented in the *phyrisk* library. The inputs are:

- The number of days per year that the daily average air temperature at the geolocation of the generation unit is above a discrete set of thresholds ranging from 25°C to 55°C. It allows to derive:
 - a discrete exceedance probability curve for air temperature in a climate-conditioned projection;
 - a design air temperature (DAT) defined as the 90%-quantile interpolated from the probability curve in a baseline scenario. It corresponds to the air temperature at which the unit generates electricity with the designed maximum efficiency.

- The disruption curve estimating the number of inoperable days in a year from the number of days per year that the air temperature reaches a threshold expressed as a number of degrees above DAT provided that it does not exceed the shutdown air temperature (SAT). SAT corresponds to the threshold above which it no longer makes technical or economic sense to keep a thermal power plant running. It is set to 50°C. The disruption curve depends on whether the generation technology is gas or steam combined with a dry cooling system. The maximum number of inoperable days interpolated for a given ‘air temperature above DAT’ threshold across potential disruption curves is chosen when turbine and/or cooling system is not provided as an asset archetype.

The output is a discrete disruption probability curve due to air temperature risks. It provides the probability in a future projection that the operations of the generation unit are disrupted for a given period expressed as a year fraction.

3.3.3.2 Water temperature

Water temperature primarily affects a power plant’s cooling efficiency and/or its ability to meet local regulatory discharge temperature standards. It has no impact on gas turbines or air-cooled units. Additionally, wet-bulb air temperature (WBGT), determined by the combined conditions of air temperature and relative humidity, affects cooling efficiency of recirculating cooling towers. The water temperature module described in [14, 15] is implemented in the *physrisk* library. The inputs are:

1. The number of weeks per year that the weekly average water temperature at the geolocation of the generation unit is above a discrete set of thresholds ranging from 5°C to 40°C. It allows to derive:
 - a discrete exceedance probability curve for water temperature in a climate-conditioned projection;
 - a design water temperature (DWT) defined as the 90%-quantile interpolated from the probability curve in a baseline scenario. It corresponds to the water temperature below which a steam turbine with a once-through cooling system does not experience generation losses.

2. The number of days per year that the daily average WBGT at the geolocation of the generation unit is above a discrete set of thresholds ranging from 5°C to 60°C. The 99%-quantile interpolated from the probability curve in a baseline scenario corresponds to the WBGT below which a steam turbine with a recirculating cooling system does not experience generation losses. The joint distribution of water temperature and WBGT in a climate-conditioned projection is estimated using a Gaussian copula with a correlation ρ set to 50% by default. The discrete exceedance probability curve for water temperature (WT) knowing that WBGT exceeds the 99%-quantile of the baseline scenario (denoted $\text{WBGT}_{\text{baseline}}^{99\%}$) is thus given by:

$$\frac{\Phi_{\rho} \left(\Phi_{\mathcal{N}(0,1)}^{-1} (\# \text{weeks/year WT} > .), \Phi_{\mathcal{N}(0,1)}^{-1} (\# \text{days/year WBGT} > \text{WBGT}_{\text{baseline}}^{99\%}) \right)}{\left(\# \text{days/year WBGT} > \text{WBGT}_{\text{baseline}}^{99\%} \right)}$$

with Φ_{ρ} the cumulative distribution functions of a ρ -correlated standard normal random pair. Scaled by the probability in a climate-conditioned projection that WBGT exceeds the 99%-quantile of the baseline scenario, it replaces the unconditional discrete exceedance probability curve calculated in the first step when the generation unit features a steam turbine with a recirculating cooling system. DWT is also capped at 24.7°C¹⁵.

3. The disruption curve estimating the number of inoperable days in a year from the number of days per year that the water temperature reaches a threshold expressed as a number of degrees above DWT. It depends on the cooling technology (once-through or recirculating) of the steam turbine. The maximum number of inoperable days interpolated for a given ‘water temperature above DWT’ threshold across potential disruption curves is chosen when turbine and/or cooling system is not provided as an asset archetype.
4. The regulatory discharge water limit estimating the number of inoperable days in a year from (intake) water temperature. It applies to steam turbine with a once-through cooling system on top of the preceding disruption impact using a max-of approach.

The output is a discrete disruption probability curve due to water temperature risks. It provides the probability in a future projection that

¹⁵ The intake water temperature equivalent to a 35°C discharge water temperature given the linear approximation:

$$T_{\text{discharge}} = 1.0191 \times T_{\text{intake}} + 9.7951^{\circ}\text{C}$$

the operations of the generation unit are disrupted for a given period expressed as a year fraction.

3.3.3.3 Heat vulnerability model

3.4 Water-related risks

3.4.1 Hazard models

Water-related risks are modelled as chronic using some indicators implemented by the World Resources Institute (WRI) Aqueduct 4.0 [13]:

- Water demand is the maximum potential water required to meet sectoral demands (domestic, industrial, irrigation and livestock).
- Water supply is the available blue water, that is, the total amount of renewable freshwater available to a sub-basin with upstream consumption removed.
- Water stress is an indicator of competition for water resources defined as the ratio of water demand by human society divided by available blue water.
- Water depletion is the ratio of total water consumption divided by available renewable water supplies. Total water consumption includes domestic, industrial, irrigation and livestock consumptive uses. Available renewable water supplies include the impact of upstream consumptive water users and large dams on downstream water availability.

3.4.2 Vulnerability models

3.4.2.1 Thermal power generation projects In high water-stress areas, decreased water availability, caused either by reduced runoff or increased demand/competition, can result in reduction in permitted water withdrawals and/or more stringent water efficiency regulations in the future. The water stress module described in [14, 15] is implemented in the *physrisk* library. The inputs are:

- The indicator for water stress in a climate-conditioned projection at the geolocation of the generation unit. A catchment with Water

Stress (WS) higher than 40% is considered highly water stressed and prone to water-stressed induced efficiency reductions impacting wet-cooled plants. In this vulnerability model, a steam turbine with a once-through or recirculating cooling system experiences generation losses when Water Stress exceeds 40%. From its mean provided as an input, Water Stress is assumed to follow a shifted uniform probability distribution

$$WS \sim (\text{water demand indicator})_{\text{climate-conditioned projection}} - 0.5 + \mathcal{U}(0, 1)$$

so that

$$\mathbb{P}(WS > 0.4) = \max\left(0, \min\left(0.1 + (\text{water demand indicator})_{\text{climate-conditioned projection}}, 1\right)\right)$$

- The indicator for water demand in both a climate-conditioned projection and a baseline scenario at the geolocation of the generation unit. It allows to calculate the Supply Reduction Rate (SRR) corresponding to the relative change of the future blue water availability from baseline

$$\text{SRR} = \frac{(\text{water demand indicator})_{\text{climate-conditioned projection}}}{(\text{water demand indicator})_{\text{baseline}}} - 1$$

- The disruption curve estimating the number of stranded days in a year from the supply reduction rate due to a limited cooling water budget. It depends on the cooling technology (once-through or recirculating) of the steam turbine. The maximum number of inoperable days interpolated for a given SRR across potential disruption curves is chosen when turbine and/or cooling system is not provided as an asset archetype.

The output is a discrete disruption probability curve due to water-related risks. It provides the probability in a future projection that the operations of the generation unit are disrupted for a given period expressed as a year fraction.

3.5 Disaggregation data sets

Where asset locations are unknown, the vulnerability of those assets to climate hazards may still be modelled by using aggregate exposure information. For example, the total value of a portfolio of assets may be known as well as the region containing those assets, even if the individual

locations are unknown. The total asset value can be *disaggregated* using a (two dimension) probability distribution of the asset locations.

For high resolution hazards such as flood and wildfire, two possible approaches are:

- Sample a number of point locations from the distribution of asset locations.
- Split the region into grid cells and integrate over the cells.

The latter is challenging in case of high resolution hazards, although techniques do exist to reduce the number of grid cells – i.e. to use a resolution lower than the resolution of the original hazard indicator data set (in cases where that may be impractical). For example, distributions of grid cell fractions experiencing hazard intensities of different magnitudes can be derived.

In this section however, we focus more on the former approach. Essentially, this is the sampling of a proxy portfolio which is used to disaggregate the total asset value and has the advantage that the proxy portfolios can be treated in the same way as known asset portfolios.

3.5.1 Sampling from two dimensional data sets

The probability density of the asset locations is given by $f_{X,Y}(x,y)$ where x and y are spatial co-ordinates. The fraction of the exposure measure (e.g. asset total value) within a particular region is given by:

$$\iint_{\text{region}} f_{X,Y}(x,y) dx dy \quad (34)$$

In order to sample n locations from $f_{X,Y}$, the marginal distribution is calculated:

$$f_X(x) = \int_y f_{X,Y}(x,y) dy \quad (35)$$

and values x_k , $k \in [1..n]$ are sampled from this. For each k , y_k is then sampled from the conditional distribution, $f_{Y|X}(y, x_k)$.

$$f_{Y|X}(y, x_k) = f_{X,Y}(x_k, y) / f_X(x_k) \quad (36)$$

In both cases, sampling is done by finding the inverse cumulative of the distribution. In the discrete case, say we have a two dimensional exposure data set $\mathbf{E}$ with elements $E_{i,j}$, i corresponding to the x spatial dimension and j the y dimension. $\sum_{i,j} E_{i,j} = 1$ by construction. We can define a discrete form of the cumulative $\int_{-\infty}^x f_X(u) du$, $F_X(i)$ as:

$$F_X(i) = \sum_{u=1}^i \sum_{j=1}^{n_j} E_{u,j} \quad (37)$$

The index i_k is then sampled by drawing uniforms p_k from $P \sim \mathcal{U}(0, 1)$ and finding i_k , the largest value of i such that $p_k < F_X(i)$

$$i_k = \sup\{i : p_k \leq F_X(i)\} \quad (38)$$

The discrete conditional is then calculated:

$$F_{Y|X}(j, i_k) = \sum_{u=1}^j E_{i_k,u} / \sum_{u=1}^{n_j} E_{i_k,u} \quad (39)$$

and index j_k is then sampled by drawing another set of uniforms q_k :

$$j_k = \sup\{j : q_k \leq F_{Y|X}(j, i_k)\} \quad (40)$$

References

- [1] ARIAS, M. E., FARINOSI, F., LEE, E., LIVINO, A., BRISCOE, J., AND MOORCROFT, P. R. Impacts of climate change and deforestation on hydropower planning in the brazilian amazon. *Nature Sustainability* 3, 6 (2020), 430–436.
- [2] BAVANDI, A., BERRAIS, D., DOLK, M., AND MAHUL, O. Physical climate risk assessment: Practical lessons for the development of climate scenarios with extreme weather events from emerging markets and developing economies. Tech. rep., Network for Greening the Financial System, 2022.
- [3] BEGUERÍA, S., VICENTE-SERRANO, S. M., REIG, F., AND LATORRE, B. Standardized precipitation evapotranspiration index (spei) revisited: parameter fitting, evapotranspiration models, tools, datasets and drought monitoring. *International journal of climatology* 34, 10 (2014), 3001–3023.
- [4] BERTRAM, C., HILAIRE, J., KRIEGLER, E., BECK, T., BRESCH, D., CLARKE, L., CUI, R., EDMONDS, J., MIN, J., PIONTEK, F., ET AL. Ngfs climate scenarios database: Technical documentation.
- [5] BLACK, F., AND COX, J. C. Valuing corporate securities: some effects of bond indenture provisions. *Journal of Finance* 31, 2 (1976), 351–367.
- [6] CHRISTIDIS, N. Using CMIP6 multi-model ensembles for near real-time attribution of extreme events. *Hadley Centre Technical Note 107* (2019).
- [7] CHRISTIDIS, N., STOTT, P. A., SCAIFE, A. A., ARRIBAS, A., JONES, G. S., COPSEY, D., KNIGHT, J. R., AND TENNANT, W. J. A new HadGEM3-A-based system for attribution of weather-and climate-related extreme events. *Journal of Climate* 26, 9 (2013), 2756–2783.
- [8] DOSIO, A., MENTASCHI, L., FISCHER, E. M., AND WYSER, K. Extreme heat waves under 1.5 C and 2 C global warming. *Environmental Research Letters* 13, 5 (2018), 054006.
- [9] DOTTORI, F., ET AL. Flood hazard maps at european and global scale, 2016.
- [10] FEMA. Fema hazus, 2025.
- [11] HUIZINGA, J., DE MOEL, H., SZEWCZYK, W., ET AL. Global flood depth-damage functions: Methodology and the database with guidelines. Tech. rep., Joint Research Centre (Seville site), 2017.

-
- [12] KENYON, C., AND BERRAHOUIA, M. Climate change valuation adjustment (ccva) using parameterized climate change impacts. *Risk* (2021).
- [13] KUZMA, S., BIERKENS, M. F., S. LAKSHMAN, S., LUO, T., SACCOCCIA, L., SUTANUDJAJA, E. H., AND VAN BEEK, R. Aqueduct 4.0: Updated decision-relevant global water risk indicators. *World Resources Institute* (2023).
- [14] LUO, T., ZHOU, L., FALZON, J., CHENG, Y., CHRISTIANSON, G., WU, Y., AND HABCHI, A. Assessing physical climate risks for the european bank for reconstruction and development’s power generation project investment portfolio.
- [15] LUO, T., ZHOU, L., FALZON, J., CHENG, Y., CHRISTIANSON, G., WU, Y., AND HABCHI, A. A framework to assess multi-hazard physical climate risk for power generation projects from publicly-accessible sources. *Communications Earth & Environment* 4, 117 (2023).
- [16] MARZI, S., MYSIK, J., ESSENFELDER, A. H., PAL, J. S., VERNACCINI, L., MISTRY, M. N., ALFIERI, L., POLJANSEK, K., MARIN-FERRER, M., AND VOUSDOKAS, M. Assessing future vulnerability and risk of humanitarian crises using climate change and population projections within the inform framework. *Global Environmental Change* 71 (2021), 102393.
- [17] MASKREY, A., PEDUZZI, P., CHATENOUX, B., HEROLD, C., DAO, Q.-H., AND GIULIANI, G. Revealing risk, redefining development, global assessment report on disaster risk reduction. *United Nations Strategy for Disaster Reduction* (2011), 17–51.
- [18] MAZDIYASNI, O., SADEGH, M., CHIANG, F., AND AGHAKOUCHAK, A. Heat wave intensity duration frequency curve: A multivariate approach for hazard and attribution analysis. *Scientific reports* 9, 1 (2019), 1–8.
- [19] MENTASCHI, L., VOUSDOKAS, M., VOUKOUVALAS, E., SARTINI, L., FEYEN, L., BESIO, G., AND ALFIERI, L. The transformed-stationary approach: a generic and simplified methodology for non-stationary extreme value analysis. *Hydrology and Earth System Sciences* 20, 9 (2016), 3527–3547.
- [20] MERTON, R. C. On the pricing of corporate debt: the risk structure of interest rates. *Journal of Finance* 29, 2 (1974), 449–470.
- [21] MITCHELL-WALLACE, K., JONES, M., HILLIER, J., AND FOOTE, M. *Natural catastrophe risk management and modelling: A practitioner’s guide*. John Wiley & Sons, 2017.

- [22] NELSEN, R. B. *An Introduction to Copulas*, 2nd ed. Princeton University Press, New York (NY), 2007.
- [23] OASIS. Oasis loss modelling framework: open source catastrophe modelling platform, 2021.
- [24] OS-C. OS-Climate (OS-C) platform, 2021.
- [25] PAPROTNY, D., AND MORALES NÁPOLES, O. O. Pan-european data sets of river flood probability of occurrence under present and future climate, 2016.
- [26] POLJANSEK, K., MARZI, S., GALIMBERTI, L., DALLA VALLE, D., PAL, J., ESSENFELDER, A., MYSIK, J., AND CORBANE, C. Inform climate change risk index. *Publications Office of the European Union* (2022).
- [27] PÖRTNER, H.-O., ROBERTS, D. C., ADAMS, H., ADLER, C., ALDUNCE, P., ALI, E., BEGUM, R. A., BETTS, R., KERR, R. B., BIESBROEK, R., ET AL. *Climate change 2022: Impacts, adaptation and vulnerability*. IPCC Geneva, Switzerland:, 2022.
- [28] PRIEST, S., ET AL. *Flood and coastal erosion risk management: handbook for economic appraisal*. 2025.
- [29] RANGER, N. A., MAHUL, O., AND MONASTEROLO, I. Assessing financial risks from physical climate shocks.
- [30] RASCHKE, M. About the return period of a catastrophe. *Natural Hazards and Earth System Sciences* 22, 1 (2022), 245–263.
- [31] REISINGER, A., HOWDEN, M., VERA, C., ET AL. *The Concept of Risk in the IPCC Sixth Assessment Report: A Summary of Cross-Working Group Discussions*. IPCC Geneva, Switzerland:, 2020.
- [32] RICHTERS, O., ET AL. Climate scenarios database: Technical documentation v3.1. Tech. rep., Network for Greening the Financial System, 2022.
- [33] SMIRNOV, O., ZHANG, M., XIAO, T., ORBELL, J., LOBBEN, A., AND GORDON, J. The relative importance of climate change and population growth for exposure to future extreme droughts. *Climatic Change* 138 (2016), 41–53.
- [34] STOTT, P. A., CHRISTIDIS, N., OTTO, F. E., SUN, Y., VANDERLINDEN, J.-P., VAN OLDENBORGH, G. J., VAUTARD, R., VON STORCH, H., WALTON, P., YIOU, P., ET AL. Attribution of extreme weather and climate-related events. *Wiley Interdisciplinary Reviews: Climate Change* 7, 1 (2016), 23–41.

- [35] TANGUY, M., PRUDHOMME, C., SMITH, K., AND HANNAFORD, J. Historical gridded reconstruction of potential evapotranspiration for the uk. *Earth System Science Data* 10, 2 (2018), 951–968.
- [36] TAYLOR, P. Calculating financial loss from catastrophes. In *SECED 2015 Conference: Earthquake risk and engineering towards a resilient world* (2015), Society for earthquake and civil engineering dynamics.
- [37] TAYLOR, P., AND CARTER, J. Oasis financial module, 2020.
- [38] THRASHER, B., WANG, W., MICHAELIS, A., MELTON, F., LEE, T., AND NEMANI, R. Nasa global daily downscaled projections, CMIP6. *Scientific Data* 9, 1 (2022), 1–6.
- [39] VICENTE-SERRANO, S., BEGUERIA, S., AND LOPEZ-MORENO, J. I. A multiscalar drought index sensitive to global warming: the standardized precipitation evapotranspiration index. *Journal of Climate* 23, 7 (2010), 1696–1718.
- [40] WARD, P. J., DE MOEL, H., AND AERTS, J. How are flood risk estimates affected by the choice of return-periods? *Natural Hazards and Earth System Sciences* 11, 12 (2011), 3181–3195.
- [41] WARD, P. J., JONGMAN, B., WEILAND, F. S., BOUWMAN, A., VAN BEEK, R., BIERKENS, M. F., LIGTVOET, W., AND WINSEMIUS, H. C. Assessing flood risk at the global scale: model setup, results, and sensitivity. *Environmental research letters* 8, 4 (2013), 044019.
- [42] WARD, P. J., WINSEMIUS, H. C., KUZMA, S., BIERKENS, M. F., BOUWMAN, A., DE MOEL, H., LOAIZA, A. D., EILANDER, D., ENGLHARDT, J., ERKENS, G., ET AL. Aqueduct floods methodology. *World Resources Institute* (2020), 1–28.
- [43] WOETZEL, J., PINNER, D., AND SAMANDARI, H. Climate risk and response: Physical hazards and socioeconomic impacts.
- [44] ZHANG, Y., AND SHINDELL, D. T. Costs from labor losses due to extreme heat in the usa attributable to climate change. *Climatic change* 164, 3 (2021), 1–18.